D. Organic Acids

Notes

- A carboxylic acid is an organic compound that contains a functional carboxyl group (C(=O)OH).
- The general formula of a carboxylic acid is R-COOH with R referring to the rest of the molecule.
- Carboxylic acids occur widely and include the amino acids (which make up proteins) and acetic acid (which is part of vinegar).
- Salts and esters of carboxylic acids are called carboxylates.
- On basis of the presence of carboxyl group in acid, carboxylic acids have been divided into three types–

1. **Monocarboxylic Acid** : contains one carboxyl group, example – acetic acid (CH_3COOH).
2. **Dicarboxylic acid :** containing two carboxyl groups, example – adipic acid (the monomer used to produce nylon).
3. **Tricarboxylic acid :** containing three carboxyl groups, example – citric acid.

- Carboxylic acids with a long aliphatic chain, which is either saturated or unsaturated are also called **fatty acids** because higher members of this group as palmitic acid ($C_{15}H_{31}COOH$) and stearic acid ($C_{17}H_{35}COOH$) were firstly derived from the fats.
- Fatty acids have even number of carbons, examples – docosahexaenoic acid and eicosapentaenoic acid (nutritional supplements).
- Fatty acids are usually not found in organisms, but instead as three main classes of esters – triglycerides, phospholipids and cholestrol esters.
- They are important dietary sources of fuel for animals and important structural components for cells.

Formic acid :

- Formic acid (HCOOH) is simplest monocarboxylic acid. It was firstly obtained by the distillation of ants (formica = ant), so its name is formic acid.
- Formic acid is an irritating chemical present in the sprayed venom of some ant species and in the secretion released from some stinging nettles.
- Formic acid is dangerous, but at low concentration it is very useful.
- Formic acid is an antibacterial substance, hence it is used as a food preservative.
- Formic acid is also used to control pest, to produce food and cosmetic additives.

Properties of Formic acid -

- Formic acid has a strong odour and is often described as having a 'pungent' smell.
- Formic acid is a colourless liquid.
- Formic acid freezes at 8.3°C and boils at 100.8°C.
- Formic acid is a **corrosive** liquid. It makes blisters on the skin.
- Formic acid is miscible with water, alcohol & ether.

Acetic Acid :

- Acetic acid (CH_3COOH) is also called ethanoic acid.
- Acetic acid is the most important acid among carboxylic acids.
- A dilute (approximately 5% by volume) solution of acetic acid produced by fermentation and oxidation of natural carbohydrates is called vinegar.
- Vinegar is sour due to presence of acetic acid.

Properties of acetic acid -

- Anhydrous acetic acid is a colourless liquid with pungent smell.
- Boiling point and melting point of acetic acid is 118°C and 16.6°C respectively.
- Acetic acid when contains a very low amount of water (less than 1%), is called anhydrous acetic acid or glacial acetic acid. The reason it is called glacial is because it solidifies into solid acetic acid crystals, at just below the room temperature (around 16-17°C).

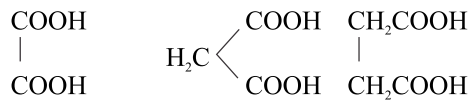
Straight chain, saturated carboxylic acids

Carbon atoms	Common name	IUPAC Name	Chemical formula	Common location or use
1	Formic acid	Methanoic acid	HCOOH	Insect stings
1	Carbonic acid	Carbonic acid	OHCOOH	Blood and tissues
2	Acetic acid	Ethanoic acid	CH_3COOH	Vinegar
3	Propionic acid	Propanoic acid	CH_3CH_2COOH	Preservative for stored grains, body odour, milk, cheese
4	Butyric acid	Butanoic acid	$CH_3(CH_2)_2COOH$	Butter

- Acetic acid is a corrosive liquid.
- Acetic acid is miscible with water, alcohol and ether.

Dicarboxylic acids :

- A discarboxylic acid is an organic compound containing two carboxyl functional groups.
- The general molecular formula for dicarboxylic acids can be written as $HO_2C–R–CO_2H$, where R can be aliphatic or aromatic.
- Dicarboxylic acids show similar chemical behaviour and reactivity to monocarboxylic acids.
- Some dicarboxylic acids are as follows :



Oxalic acid **Malonic acid** **Succinic acid**

Question Bank

1. Assertion (A) : Formic Acid is a stronger acid than acetic acid.
Reason (R) : Formic Acid is an organic acid.
Code :
(a) Both (A) and (R) are true and (R) is the correct explanations of (A).
(b) Both (A) and (R) are true but (R) is not correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

I.A.S. (Pre) 1998

Ans. (b)

Formic acid (HCOOH) and acetic acid (CH_3COOH) both are organic acids. Acetic acid contains a electron donating methyl group (CH_3). While formic acid has no such electron donating group and loss of proton (H^+) in it is comparatively easy and hence it is stronger acid than acetic acid.

2. Which acid is described as HOOCCOOH?
(a) Carbonic acid
(b) Oxalic acid
(c) Acetic acid
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

Oxalic acid is a dicarboxylic acid with a chemical formula $C_2H_2O_4$ also written as $(COOH)_2$ of $(CO_2H)_2$. It is also known as Ethanedioic acid or Oxiric acid. This organic compound is found in many vegetables and plants. It is the simplest dicarboxylic acid with condensed formula HOOC – COOH and has an acidic strength greater than acetic acid.

3. The sensation of fatigue in the muscles after prolonged strenuous physical work is caused by–
(a) A decrease in the supply of oxygen
(b) Minor wear and tear of muscle fibres
(c) The depletion of glucose
(d) The accumulation of lactic acid

I.A.S. (Pre) 2000

Ans. (d)

After hard physical work, due to the accumulation of lactic acid [$CH_3CH(OH)COOH$] in muscles, the body would experience fatigue. The rapid accumulation of lactic acid is not a temporal process. Rest is needed to remove tiredness. A body massage also helps to get relief from fatigue.

4. Accumulation of which one of the following in the muscles leads to fatigue?
(a) Lactic acid (b) Benzoic acid
(c) Pyruvic acid (d) Uric acid

U.P. P.C.S. (Pre) 2010
U.P. Lower Sub. (Spl.) (Pre) 2008
U.P.P.C.S. (Pre) 1992

Ans. (a)

See the explanation of above question.

5. Which acid accumulates in the muscles to cause fatigue?
(a) Lactic acid (b) Pyruvic acid
(c) Citric acid (d) Uric acid
(e) Acetic acid

Chhattisgarh P.C.S (Pre) 2013

Ans. (a)

See the explanation of above question.

6. Lemon is citrus due to –
(a) Hydrochloric acid (b) Acetic acid
(c) Tartaric acid (d) Citric acid

39th B.P.S.C. (Pre) 1994

Ans. (d)

Lemon contains mainly citric acid ($C_6H_8O_7$) which fulfils the deficiency in the body. Citric acid is a weak organic acid found in citrus fruits. Citric acid is most concentrated in lemons and limes, where it can comprise as much as 8% of the dry weight of the fruit. Acetic acid is found in vinegar, while tartaric acid is found in tamarind.

7. Which acid is mainly find in lemons?
(a) Acetic acid (b) Ascorbic acid
(c) Citric acid (d) Nitric acid

M.P.P.C.S. (Pre) 2003

Ans. (c)

See the explanation of above question.

8. Match List-I and List-II and select the correct answer from the code given below :

List-I (Acid)	**List-II (Source)**
A. Lactic acid	**1. Lemon**
B. Acetic acid	**2. Rancid butter**
C. Citric acid	**3. Milk**
D. Butyric acid	**4. Vinegar**

Code :

	A	B	C	D
(a)	1	4	3	2
(b)	3	1	4	2
(c)	2	3	4	1
(d)	3	4	1	2

U.P.P.C.S. (Pre) 1997

Ans. (d)

Lactic acid is found in milk, acetic acid is found in vinegar, citric acid is found in lemon and butyric acid is found in rancid butter.

9. Match List-I with List-II and select the correct answer from the code given below the lists :

List-I	**List-II**
A. Acetic Acid	**1. Butter**
B. Lactic acid	**2. Lemon**
C. Butyric acid	**3. Vinegar**
D. Citric acid	**4. Milk**

Code :

	A	B	C	D
(a)	3	4	2	1
(b)	3	4	1	2
(c)	4	3	1	2
(d)	1	2	3	4

U.P. Lower Sub. (Mains) 2013

Ans. (b)

See the explanation of above question.

10. Which one of the following pairs is not correctly matched?

(a) Lactose – Sour milk
(b) Carbonic acid – Soda water
(c) Formic acid – Red ants
(d) Tartaric acid – Grape juice

U.P. Lower Sub. (Pre) 2015

Ans. (a)

Lactic acid is found in milk not the lactose. Lactose is a milk sugar. Lactic acid is produced by the fermentation of lactose. Other three pairs are correctly matched.

11. Which one of the following pairs is not correctly matched?

(a) Ascorbic acid – Lemon
(b) Maltose – Malt
(c) Acetic acid – Curd
(d) Formic acid – Red Ant

U.P.P.C.S. (Mains) 2015

Ans. (c)

Ascorbic acid or vitamin C is found in many fresh vegetables and fruits such as cauliflower, lemon, cabbage and citrus fruits. Maltose is found in Malt. Malt is a germinated cereal that has been dried in a process known as "Malting". Lactic acid is found in curd not acetic acid. Formic acid occurs in the body of red ants and in the stings of bees.

12. Match List-I with List-II and select the correct answer by using the codes given below the lists :

List-I	**List-II**
A. Pickle	**1. Carbonic Acid**
B. Sour Milk	**2. Acetic Acid**
C. Apple	**3. Lactic Acid**
D. Cold drinks and soda water	**4. Malic Acid**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	4	1
(c)	4	3	1	2
(d)	3	4	2	1

U.P.P.C.S. (Mains) 2003

Ans. (b)

The correctly matched lists are as follows :

Pickle	–	Acetic Acid
Sour Milk	–	Lactic Acid
Apple	–	Malic Acid
Cold Drinks and soda water	–	Carbonic Acid

13. Which of the following is NOT correctly matched ?

(a) Acid present in vinegar – Acetic acid
(b) Compound present in bones – Calcium phosphate
(c) Souring of milk – Nitric acid
(d) Acid present in gastric juice – Hydrochloric acid

U.P. P.C.S. (Pre) 2021

Ans. (c)

Soured milk denotes a range of food products produced by the acidification of milk. It is the lactic acid which makes the milk sour. Pairs of other options are correctly matched.

14. Milk openly placed for sometime becomes sour due to-

(a) Carbonic acid (b) Lactic acid

(c) Citric acid (d) Acetic acid

U.P. Lower Sub. (Pre) 2004

Ans. (b)

Milk contains a sugar called lactose. It also contains harmless bacteria called lactobacillus, which uses glucose for energy and creates lactic acid as a by-product. It is the lactic acid which makes the milk sour. The presence of lactic acid or lactate in milk is due to the fermentation of lactose caused mainly by lactic bacteria.

15. Which one of the following acids is present in sour milk products ?

(a) Acetic Acid (b) Butyric Acid
(c) Tartaric Acid (d) Lactic Acid

U.P.P.C.S. (Mains) 2012

Ans. (d)

See the explanation of above question.

16. Which one of the following acids, is formed during the change of milk into curd?

(a) Acetic acid (b) Ascorbic acid
(c) Citric acid (d) Lactic acid

R.A.S./R.T.S.(Pre) 2008

Ans. (d)

When pasteurized milk is heated to a temperature of 30-40^{o}C or even at room temperature and a small amount of old curd or whey is added to it, the lactobacillus (bacteria) in that curd or whey sample starts to grow. They convert the lactose into lactic acid, which imparts the sour taste to curd.

17. Curd making is an ancient "Biotechnological" process involving :

(a) Bacteria (b) Virus
(c) Fungus (d) Protozoa

R.A.S./R.T.S.(Pre) 2010

Ans. (a)

See the explanation of above question.

18. The predominant organic acid in grapes is –

(a) Acetic acid (b) Citric acid
(c) Malic acid (d) Tartaric acid

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (d)

Tartaric acid is the predominant organic acid in grapes. It is a white crystalline diprotic acid.

19. For human nutrition, tomatoes are a rich source of

(a) Acetic acid (b) Methonic acid
(c) Citric acid (d) Oxalic acid

R.A.S./R.T.S. (Re. Exam) (Pre) 2013

Ans. (c)

Tomato contains mainly citric acid and malic acid. Due to which, it also acts as ant-acid. It also contains oxalic acid which is found less in quantity.

20. Which one of the following organic acids is abundant in grapes, tamarind and banana?

(a) Acetic acid (b) Citric acid
(c) Lactic acid (d) Tartaric acid

U.P.P.C.S. (Pre) 2017

Ans. (d)

Tartaric acid is found in abundance in grapes, tamarind and banana. This acid is used as regulator and antioxidant in food items.

21. Which acid is used in photography–

(a) Formic acid (b) Oxalic acid
(c) Citric acid (d) Acetic acid

U.P.U.D.A./L.D.A. (Pre) 2003

Ans. (b)

Oxalic acid is used in photography. Formic acid is found in ants. Citric acid is found in lemon and acetic acid is found in vinegar.

22. Which one of the following acids is used in the manufacturing of baking powder ?

(a) Oxalic Acid (b) Lactic Acid
(c) Tartaric Acid (d) Benzoic Acid

U.P.P.C.S. (Mains) 2012

Ans. (c)

Tartaric acid is used in the manufacturing of baking powder. This tartaric acid occurs naturally in many plants particularly in grapes, bananas and tamarinds. It is commonly combined with baking soda to function as a leavening agent in recipes and is one of the main acids found in wine.

23. The odour of acetic acid resembles that of :

(a) Vinegar
(b) Tomato
(c) Kerosene
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

The odour of acetic acid (CH_3COOH) resembles that of vinegar. Vinegar is a liquid consisting mainly of acetic acid and water.